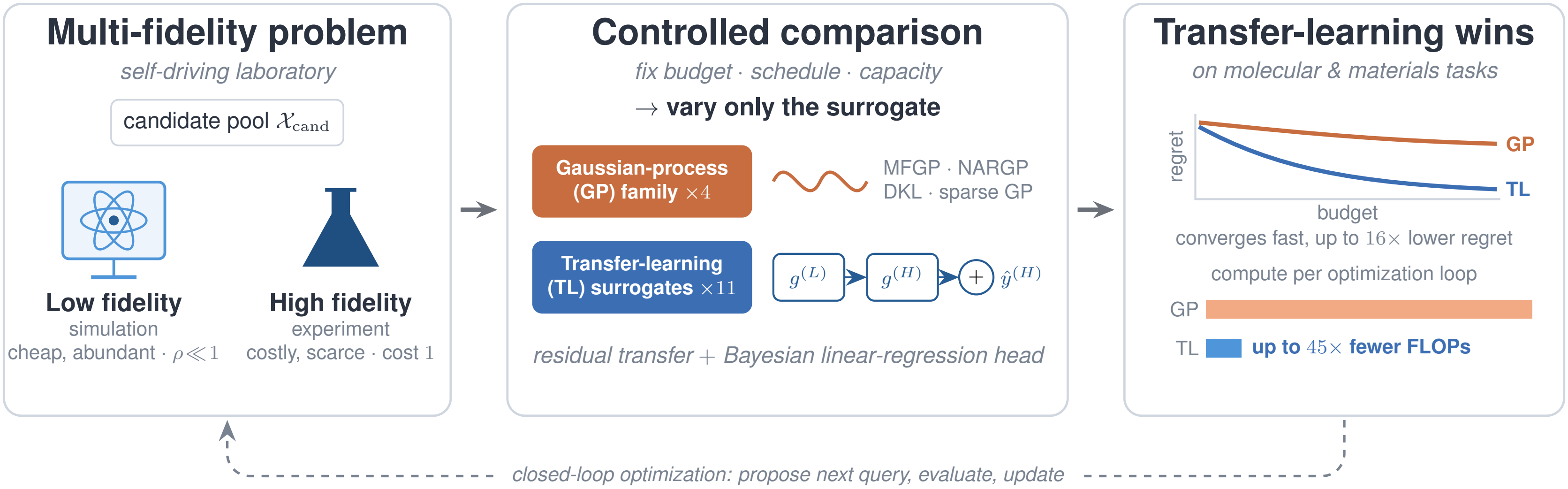
Multi-fidelity problem
self-driving laboratory
candidate pool $\mathcal{X}_{\text{cand}}$
Low fidelity
simulation
cheap, abundant · $\rho \ll 1$
High fidelity
experiment
costly, scarce · cost 1
Controlled comparison
fix budget · schedule · capacity
$\rightarrow$ vary only the surrogate
Gaussian-process (GP) family $\times 4$
MFGP · NARGP
DKL · sparse GP
Transfer-learning (TL) surrogates $\times 11$
$g^{(L)}$
$g^{(H)}$
$+$
$\hat{y}^{(H)}$
residual transfer $+$ Bayesian linear-regression head
Transfer-learning wins
on molecular & materials tasks
regret
GP
TL
budget
converges fast, up to $16\times$ lower regret
compute per optimization loop
GP
TL
up to $45\times$ fewer FLOPs
closed-loop optimization: propose next query, evaluate, update